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Integrated optimization of railway freight operation planning and pricing based on carbon emission reduction policies

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ABSTRACT

Cargo transportation is one of the major sources of carbon emissions. To reduce carbon emissions from inland freight, one way is to encourage rail transportation instead of truck transportation. In this paper, a new approach combining railway operation management, revenue management, and policy formulation is proposed to increase the market share of railway freight and reduce carbon emissions. Through the integrated optimization of operation planning and pricing, the competitiveness of railway freight services is improved. The carbon tax policy acting on the demand-side can affect consigners' choice behavior. The low-carbon subsidy policy acting on the supply-side can encourage rail transport. Then a bi-level multi-objective model is proposed to achieve the trade-off among the three stakeholders (the government, rail service operators, and consigners). The complexity of the bi-level multi-objective problem motivates the development of a hybrid optimization algorithm that combines the Non-dominated sorting genetic algorithm III (NSGA-III) and the Descent algorithm based on sensitivity analysis (SAB). Through a real case based on a rail service network with nine nodes in China, the rationality of the proposed methodology is verified. The results show that the approach proposed can achieve a 39.27% modal shift (MS) from road to rail, as well as a 37.09% carbon emission reduction.
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1. Introduction

Transportation activities generated 24% of the world's carbon dioxide emissions from fuel combustion, which is the second-largest contributor behind electricity and heat generation (IEA, 2017). There are two possible reasons. One is due to the high market share of road transport. Freight transport in China has grown by almost 82% over the last decade. Approximately 76% of them depend on road haulage (NBS, 2019). Road freight market share in the EU is about four times that of railways (Kaack et al., 2018). Trucking is the dominant freight mode in the United States, carrying nearly three-quarters of the annual tonnage transported (Williams et al., 2017). This extremely unbalanced mode split of freight transport results in negative impacts of transportation. The other is that the energy consumption per unit turnover of road transportation is often several times that of railway transportation (Garcia-Alvarez et al., 2013). Lin et al. (2017) founded that the total carbon emissions of highway freight transport are 7.89–10.45 times more than that of railway freight transport. If one ton-kilometer workload can be shifted from highway to railway, the carbon emissions will reduce 38.87 g. Rail transportation is considered to be more environment-friendly than road transportation (Kim and Van Wee, 2009). Modal shift (MS) from road to rail makes it possible to reduce the transportation costs and carbon emissions, which is of great significance for realizing the national energy conservation and emission reduction strategy.

MS has been valued in many countries. For many years, the EU has been promoting the shift from road freight to more sustainable modes. In 2011, the Transport White Paper issued by the EU stated that by 2030, 30% of long-distance (more than 300 km) road freight should be shifted to other modes, such as railway or waterway, and this ratio should be higher than 50% by 2050 (EU, 2011). The EU has stepped up investment in railway transportation systems, and the Shift2Rail project is being implemented to promote the development of railway technology and model shift strategies (Shift2Rail, 2020). Argonne National Laboratory in the United States

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Table 1
Summary table of abbreviations.

Abbreviation	Explanation
MS	Modal shift
NSGA-III	Non-dominated sorting genetic algorithm III
SAB	Descent algorithm based on sensitivity analysis
IEA	International Energy Agency
NBS	National Bureau of Statistics of China
EU	European Union
MNR	Ministry of Natural Resources of China

summarized the evaluation of the potential reduction in energy use and greenhouse gas emissions achieved by shifting freight from truck to rail in a 2017 report. They estimated that by 2040, 4.1% of trucking activities could be transferred to railways (Zhou et al., 2017). In July 2001, Japan issued the "New Comprehensive Program of Logistics Policies" to achieve a 50% share of rail and water by 2010 (Horn and Nemoto, 2005). The Government of India aims to make equal shares for road and rail by 2032 (India, 2014). In 2018, the General Office of the State Council of China issued the "Three-Year Action Plan for Advancing the Adjustment of Transport Structure (2018–2020)", which pointed out the MS direction of "road to rail, road to water" (China, 2018). The main goal of this action plan is to significantly optimize the structure of China's freight transportation by 2020. More specifically, from 2018 to 2020, China aims to increase its rail freight volume by 1.1 billion tons and to reduce the road freight volume linked to coastal ports by 440 million tons.

About how to promote MS from road to rail,;Islam et al. (2016) proposed the following three critical questions: How can rail improve the service quality to attract customers? How can rail offer its customers a competitive price with road? How can rail improve its ability to cope with the increasing demand due to MS? The first two questions will be investigated in this paper. For the third question, Lin et al. (2017) have studied the expansion of the existing railway network to enable the railway to manage the highway's freight. This paper proposes a new approach to promote MS from road to rail through the implementation of carbon emission reduction policies and the optimization of railway transportation network at an operational level. The government intends to adopt the following two policies: subsidizing rail freight transportation, internalizing environmental costs. The service competitiveness of rail freight will be improved through integrated optimization of operation planning and pricing.

The remaining parts of this paper are organized as follows. Section 2 briefly reviews the existing literature. In Section 3, we focus on the model establishment. Section 4 gives the algorithm to solve the optimization problem. Section 5 is the empirical analysis, followed by conclusions in Section 6. A list of the abbreviations used in this paper is provided in Table 1.

2. Literature review

To achieve the goal of decarbonization, several measures have been proposed, such as clean vehicle technology (McCollum et al., 2018), optimizing network (Lam and Gu, 2016), and MS (Tao et al., 2017).

Jonkeren et al. (2019) presented a literature overview of research methods that evaluated the effectiveness of MS. It including choice modeling (Regmi and Hanaoka, 2015), life cycle assessment (Nocera and Cavallaro, 2016), strategic freight transport network models (Mostert et al., 2017). The effectiveness of policy measures to promote MS is a hot topic (Behrends, 2017; Blauwens et al., 2006; Bouchery and Fransoo, 2015). Choi et al. (2014) examined the impact of policy measures (including subsidy, tax, pricing, services, infrastructure, strategic planning, and operation) on promoting MS through statistical analysis. Also, the interactions among these policy measures were analyzed. The results showed that services have the most significant impact on MS, followed by infrastructure, subsidy, tax, and pricing. Moreover, subsidy, tax, and pricing have a direct effect on services. The study of Kaack et al. (2018) showed that MS could be promoted by policies targeting infrastructure investments and internalizing external costs of road freight, but few countries have such policies. The carbon tax is a way of internalizing external costs, and its rational application can promote MS (Ramseur et al., 2012). The research of Tao (2013) aimed at sea-rail intermodal transport. It indicated that subsidy policy could indirectly reduce shipper's transport costs, and thus became an effective means for MS.

One of the important tools needed to ensure the sustainable development of railway transportation is the liberalization of the railway freight market (Dolinayova et al., 2016). The introduction of competition mechanisms in the transportation industry can stimulate market vitality. The "invisible hand" of the market mechanism regulates and allocates transportation resources to achieve optimal utilization. Deregulation and market liberalization have been the main policy options chosen by the EU to promote rail freight. However, rail freight still faces imperfect competition, whether it is the competition between modes of transportation (such as road transport and rail transport) or internal competition between railway operators (Crozet, 2017). The heavily regulated railroads in the United States were largely deregulated by the 4R Act and Staggers Rail Act in 1980, which made them more dependent on the free market to promote railroad profitability (Grimm and Pittman, 2018). How to improve the competitiveness of the more environmentally-friendly mode (rail freight) in the free market is vital for sustainable transportation. In the logistics and transportation industry, the improvement of service quality is an essential means to enhance competitiveness (Islam and Eidhammer, 2016; Zhang et al., 2017). China implemented railway freight reform in 2013. One of its main contents is the gradual deregulation of transportation pricing power, which has given China's railways the right to participate in free-market competition. Establishing a price mechanism that meets fluctuating demand in the free market can enhance the competitiveness of rail. The joint optimization of operation planning and pricing can adjust the fluctuating demand of the free market to adapt to the transportation supply, which is an excellent option to avoid waste of transportation resources and make the railway profitable.

Firstly, the optimization of rail operation plan is an effective way to improve the service quality. Rail operations include complex and interrelated planning and execution decisions, such as car blocking, train scheduling and routing, resource allocation, and empty car repositioning. The train routing problem (Sama et al., 2016, 2017) was usually studied together with the scheduling problem (Fang et al., 2017; Sun et al., 2014). Timetable design must rely on accurate computations of realizable train paths and buffer times. Besinovic et al. (2016) provided an integrated micro-macro approach to study it. For the resource allocation problem, Suyabatmaz and Sahin (2015) studied a crew capacity planning problem at the tactical level. Janacek et al. (2017) proposed a new method to solve the periodic crew scheduling problem, and the example showed that it was suitable for practical application. Other train operation problems, such as the empty car repositioning problem (Bektas et al., 2009; Upadhyay and Bolia, 2014), yard operations planning (Shi and Zhou, 2015), have also been studied. It can be seen that operations of railway freight are hugely complicated. The operation plan studied in this paper is a tool to realize the transformation of the cargos flow to the train flow, and the

following aspects need to be determined: (1) the train operating route; (2) the frequency of service; (3) the number of cars grouped within trains; (4) the transfer operation plan.

Secondly, the pricing strategy is an essential part of revenue management. In rail transport, revenue management is mainly used for passenger transport (Bharill and Rangaraj, 2008; Itaru, 2007; Sibdari et al., 2008). Armstrong and Meissner (2010) overviewed the application of revenue management to rail freight and passenger transport activities. Weatherford and Bodily (1992), Harker and Hong (1994), and Talluri and Van Ryzin (2006) analyzed the favorable conditions for the realization of revenue management in railway freight transport. Strasser (1996) pointed out that railway transportation activities should give full play to the leverage of transportation prices and maximize the benefits of railway transportation activities through the dynamic relationship between price and demand. The pricing strategy of the Burlington Northern Santa Fe Railway Company (BNSF) in multimodal transport was studied by Gorman (Gorman, 2005, 2015; Gorman and Harrod, 2010). Xu et al. (2018) studied the dynamic pricing method of high-speed railway passenger transport.

The analysis of an integrated approach combining operations and revenue management is an interesting perspective of research in an attempt to study the connections between network planning and pricing policies. Li and Tayur (2005) developed a mathematical programming model for multimodal transport that considered both pricing and operational planning. Robenek et al. (2018) extended the problem raised by the literature Robenek et al. (2016), which combines passenger-centric train schedules with fare issues, while revenue management in the existing literature is performed on a fixed schedule. Due to the complexity of the rail freight system, this integration has not been considered frequently in the literature.

Based on prior research, this work makes the following contributions by jointly considering the carbon emission reduction policies, optimization of rail freight network at an operational level, and customer choice behavior. Firstly, a bi-level multi-objective mixed-integer programming model is formulated to capture the dominating relationship among the three stakeholders involved in the situation (the government, the rail freight operators, and consigners). Then we design a hybrid algorithm that combines NSGA-III, SAB algorithm, and CVX toolbox in MATLAB to solve the optimization problem. Secondly, we focus on a combination of rail operational management (operational level), revenue management (tactical level), and carbon emission reduction policies (strategic level) to promote MS from road to rail. Meanwhile, the operation planning and pricing in this paper is aimed at a more complex situation of transportation service with transshipments, which is more practical. Thirdly, the solutions under different policy factors are compared, which provides further managerial insights regarding the carbon emission reduction in transportation.

3. Mathematical formulation

3.1. Problem description

In this paper, we will study the joint optimization decision of railway freight operation plan and freight rate under a series of carbon emission reduction policies formulated by the government. Note that this is a complex problem, in part because of the need to capture the decisions of the government, rail operators, and consigners.

The government aims to maximize the reduction of carbon emissions generated by transportation activities and to minimize the cost of promoting this goal. In this paper, the subsidies paid to rail operators are used to measure the government's costs. Rail freight operators need to develop a plan to transport cargos from the origins to the destinations and maximize transportation profits. Consigners choose the transportation mode with minimal cost. The cost considered by consigners includes freight rate, transportation time, convenience, and carbon tax levied by the government to internalize the environmental cost of transportation.

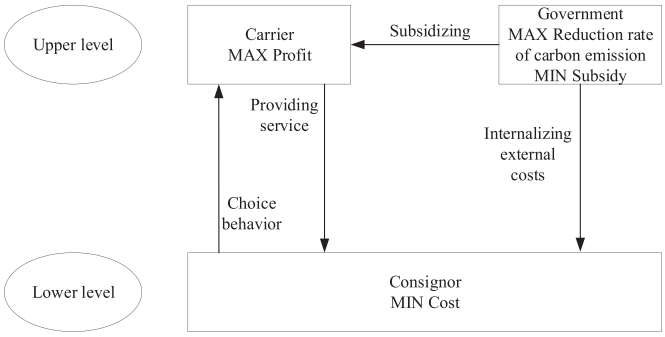


Fig. 1. Schematic view of the bi-level multi-objective model.

A bi-level multi-objective model is used to address the problem. The objectives of the government and rail operators are achieved in the upper-level model. The aim of consigners is achieved in the lower-level model. Fig. 1 depicts the schematic view of it, which indicates the interactions between the two levels. The parameters and variables used for modeling are listed in Table 2.

This paper considers a transport network (illustrated in Fig. 2) formed by three layers of departure stations $\mathscr{O} = \{O_1, O_2, \cdots, O_K\}$, intermediate stations $\mathbb{S} = \{S_1, S_2, \cdots, S_J\}$, and terminal stations $\mathscr{D} = \{D_1, D_2, \cdots, D_K\}$. In this transportation network, K trains ($O_1 \rightarrow D_1$, $O_2 \rightarrow D_2, \cdots, O_K \rightarrow D_K$) are arranged. The freight demands of $S_j \rightarrow D_k$ are met by train k ($O_k \rightarrow D_k$). The rail network in this study is unidirectional.

3.2. Upper level: carbon tax and subsidy

If all of the known total freight demands $\boldsymbol{B} = \sum_{i \in \mathbb{N}} \sum_{j \in \mathscr{D}} B_{ij}$ are satisfied by road, then the carbon emissions can be calculated as equation (1).

$$E_{road-only} = \sum_{i \in \mathbb{N}} \sum_{j \in \mathscr{D}} \sigma^{road} l_{ij}^{road} B_{ij} \tag{1}$$

The government expects to shift part of cargos to rail to reduce the carbon emissions generated by transportation activities. Carbon emissions of the transport system after the shift can be calculated according to equation (2), which is obtained by adding emissions generated by rail and road freight. Equation (3) gives the emission reduction rate due to the shift.

$$E_{road-rail} = \sum_{i \in \mathbb{N}} \sum_{j \in \mathscr{D}} \sigma^{road} l_{ij}^{road} q_{ij}^{road} + \sum_{i \in \mathbb{N}} \sum_{j \in \mathscr{D}} \sigma^{rail} l_{ij}^{rail} q_{ij}^{rail} \tag{2}$$

$$\mu(\boldsymbol{q}) = \frac{E_{road-only} - E_{road-rail}}{E_{road-only}} \tag{3}$$

The government subsidizes rail transport at a standard λ yuan per ton of carbon emission to promote this MS, as shown in equation (4).

$$U(\boldsymbol{q}) = \sum_{i \in \mathbb{N}} \sum_{j \in \mathscr{D}} \lambda \sigma^{rail} l_{ij}^{rail} q_{ij}^{rail} \tag{4}$$

Therefore, the objectives of the government can be described as

Table 2
Summary of notation.

Sets:	
$\mathscr{O}$	Set of departure stations. $\mathscr{O} = \{O_1, O_2, \cdots, O_K\}$
$\mathscr{D}$	Set of terminal stations. $\mathscr{D} = \{D_1, D_2, \cdots, D_K\}$
$\mathbb{S}$	Set of intermediate stations. $\mathbb{S} = \{S_1, S_2, \cdots, S_J\}$
$\mathscr{K}$	Set of trains. $\mathscr{K} = \{1, 2, \cdots, K\}$
$\mathbb{N}$	$\mathbb{N} = \mathscr{O} \cup \mathbb{S}$
$\mathscr{W}$	Set of arcs in the rail transportation network. $\mathscr{W} = \{1, 2, \cdots, W\}$
$\mathscr{M}$	Set of transportation mode. $\mathscr{M} = \{1, 2\}$, where "1" represents rail freight and "2" represents road freight
Parameters:	
B_{ij}	Freight demand from $i \in \mathbb{N}$ to $j \in \mathscr{D}$ (ton)
$\sigma^{rail}, \sigma^{road}$	Carbon emissions per ton-km on rail (σ^{rail}) and road (σ^{road}) (ton/ton-km)
$l_{ij}^{rail}, l_{ij}^{road}$	Rail mileage (l_{ij}^{rail}) and road mileage (l_{ij}^{road}) between $i \in \mathbb{N}$ and $j \in \mathscr{D}$ (km)
λ	Carbon subsidy coefficient (yuan/ton carbon emissions)
η	Carbon tax rate (yuan/ton carbon emissions)
G_{teu}	The weight of a 20-foot equivalent unit (ton/TEU)
c_{fix}^k	Fixed operating cost of train $k \in \mathscr{K}$ (yuan)
c_{var}^k	Variable operating cost of train $k \in \mathscr{K}$(yuan/TEU)
$c_{transfer}$	Unit transfer operation cost (yuan/ton)
$R_k^{\max}$	Capacity (number of cars) of train $k \in \mathscr{K}$
$q_i^{\max}$	The maximum service capability of station $i \in \mathbb{N}$ (ton)
A	A positive number with a large value
$g_{w,k}$	Incidence matrix: $g_{w,k} = 1$ if the train $k \in \mathscr{K}$ occupies the transport arc $w \in \mathscr{W}$, otherwise, $g_{w,k} = 0$
ξ_w	The maximum number of trains allowed to pass at the same time on transport arc $w \in \mathscr{W}$
$p_{ij,\min}^{rail}, p_{ij,\max}^{rail}$	Finite constants which denote lower ($p_{ij,\min}^{rail}$) and upper ($p_{ij,\max}^{rail}$) bounds for the rail freight price from $i \in \mathbb{N}$ to $j \in \mathscr{D}$ (yuan)
Decision variables:	
$q_{ij,k}^{rail}$	Rail freight volume from $i \in \mathbb{N}$ to $j \in \mathscr{D}$ carried by train $k \in \mathscr{K}$ (ton)
q_{ij}^{road}	Road freight volume from $i \in \mathbb{N}$ to $j \in \mathscr{D}$ (ton)
$\phi_{S_j}^{O_k D_h}$	Binary indicator: $\phi_{S_j}^{O_k D_h} = 1$ if the train $k \in \mathscr{K}$ unloads the cargos from O_k to D_h, $h \neq k$ at intermediate station $S_j \in \mathbb{S}$, otherwise, $\phi_{S_j}^{O_k D_h} = 0$
ψ^k	Service frequency of train $k \in \mathscr{K}$ during the decision period
m^k	Number of cars in train $k \in \mathscr{K}$ (2 TEU in each car)
p_{ij}^{rail}	Rail freight price between $i \in \mathbb{N}$ and $j \in \mathscr{D}$ (yuan/ton)

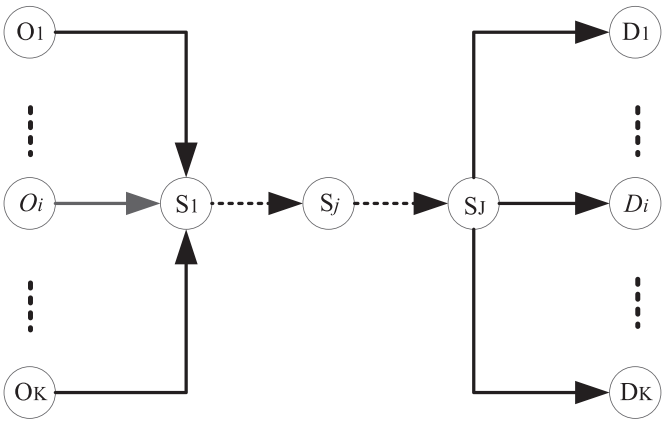


Fig. 2. Transportation network.

follows:

$$\textbf{max}\ \mu(\boldsymbol{q}) \tag{5}$$

$$\min U(\boldsymbol{q}) \tag{6}$$

If the environmental burden of transportation is not directly related to the consignors' interests, it is usually not considered when selecting the mode of transport. Carbon tax policy has been investigated in the literature intensively as a means to reduce carbon emissions. If the government sets a carbon tax with the rate of η yuan per ton of carbon emission, then the increased unit transportation cost of consigners can be calculated according to equation (7).

$$r_{ij}^m = \eta \sigma^m l_{ij}^m, m \in \mathscr{M} \tag{7}$$

3.3. Upper level: rail freight operation planning and pricing

The objective of railway operators is to maximize transportation profits, which can be described as follows:

$$\max\ I(\boldsymbol{p}, \boldsymbol{q}) - C(\boldsymbol{q}, \boldsymbol{m}, \boldsymbol{\psi}) + U(\boldsymbol{q}) \tag{8}$$

where $I(\boldsymbol{p}, \boldsymbol{q})$ and $C(\boldsymbol{q}, \boldsymbol{\psi}, \boldsymbol{\tau}, \boldsymbol{\phi})$ represent transportation revenue and cost, respectively.

$$I(\boldsymbol{p}, \boldsymbol{q}) = \sum_{i \in \mathbb{N}} \sum_{j \in \mathscr{D}} p_{ij}^{rail} q_{ij}^{rail} \tag{9}$$

$$C(\boldsymbol{q}, \boldsymbol{m}, \boldsymbol{\psi}) = \sum_{k=1}^{K} \left(c_{fix}^k + c_{var}^k m^k \right) \psi^k + c_{transfer} \sum_{k,h=1, h \neq k}^{K} q_{O_k D_h}^{rail} \tag{10}$$

$\sum_{k=1}^{K} (c_{fix}^k + c_{var}^k m^k) \psi^k$ is the train operating cost. $c_{transfer} \sum_{k,h=1,h\neq k}^{K} q_{O_k D_h}^{rail}$ is the transfer cost, where $\sum_{k,h=1,h\neq k}^{K} q_{O_k D_h}^{rail}$ is the freight volume that needs to be transferred in the intermediate stations.

The formulation with the objective function defined by formula (8) also includes the following constraints:

(1) Constraints of train operating

The service frequency of the train $k \in \mathscr{K}$ is determined by the section with the highest traffic density along the route. The freight volume of train $k \in \mathscr{K}$ at the station S_j can be calculated as equation (11).

$$q_k^{S_j} = q_{O_kD_k}^{rail} + q_{S_jD_k}^{rail} + \sum_{h=1,h\neq k}^{K} q_{O_hD_k}^{rail}\phi_{S_j}^{O_hD_k} - \sum_{h=1,h\neq k}^{K} q_{O_kD_h}^{rail}\phi_{S_j}^{O_kD_h} \tag{11}$$

where $q_{O_kD_k}^{rail}$ is the freight volume from O_k to D_k, $q_{S_jD_k}^{rail}$ is the freight volume from S_j to D_k, $\sum_{h=1,h\neq k}^{K} q_{O_hD_k}^{rail}\phi_{S_j}^{O_hD_k}$ is the freight volume unloaded by train $h \neq k$ that needs to be sent to D_k at station S_j, $\sum_{h=1,h\neq k}^{K} q_{O_kD_h}^{rail}\phi_{S_j}^{O_kD_h}$ is the freight volume unloaded by train $k \in \mathscr{K}$ that needs to be sent to $D_h, h \neq k$ at station S_j.

The freight train can only operate if its load reaches a certain proportion $\rho_{\min}$. Therefore, inequality (12) needs to be satisfied.

$$\rho_{\min} \leq \frac{q_k}{\psi^k m^k G_{Teu}} \leq 1, \quad \forall k \in \mathscr{K} \tag{12}$$

$$0 < m^k \leq R_k^{\max} \tag{13}$$

where $q_k = \max(q_k^{S_j}, \sum_{h=1}^{K} q_{O_kD_h}^{rail})$. Inequality (13) indicates that the number of cars of train $k \in \mathscr{K}$ cannot exceed its capacity $R_k^{\max}$.

(2) Capacity constraints of stations

The stations have limited processing capacity, therefore,

$$0 \leq \sum_{k,h=1}^{K} q_{O_kD_h}^{rail} \leq q_{O_k}^{\max} \tag{14}$$

$$0 \leq \sum_{k=1}^{K}\sum_{h=1,h\neq k}^{K} q_{O_kD_h}^{rail}\phi_{S_j}^{O_kD_h} \leq q_{S_j^1}^{\max} \tag{15}$$

$$0 \leq \sum_{k=1}^{K} q_{S_jD_k}^{rail} + \sum_{k=1}^{K}\sum_{h=1,h\neq k}^{K} q_{O_hD_k}^{rail}\phi_{S_j}^{O_hD_k} \leq q_{S_j^2}^{\max} \tag{16}$$

where $q_{S_j^1}^{\max}$ and $q_{S_j^2}^{\max}$ represent the maximum service capacity of the arrival area and departure area of station S_j, respectively.

(3) Capacity constraints of railway sections

The number of trains on the rail line should not exceed the capacity limit of each transport arc, therefore,

$$\sum_{k=1}^{K} g_{w,k}\psi^k \leq \xi_w, \forall w \in \mathscr{W} \tag{17}$$

(4) Constraints of transfer uniqueness

In railway freight transportation, all the cargos with the same origin and destination are regarded as an integral whole, and only one service route can be selected in the service network. Specifically, cargos that cannot be transported directly from origin to destination (e.g., cargos from station O_k to station $D_h (h \neq k)$) are unloaded at and only at one intermediate station, and then taken away by a train passing through this intermediate station and terminating at the same destination. Namely, the following step function exists:

$$\sum_{S_j \in \mathbb{S}} \phi_{S_j}^{O_kD_h} = \begin{cases} 1, q_{O_kD_h}^{rail} > 0, \forall k, h \in \mathscr{K}, h \neq k \\ 0, otherwise \end{cases} \tag{18}$$

Constraint (18) can be linearized to facilitate model solving:

$$A\sum_{S_j \in \mathbb{S}} \phi_{S_j}^{O_kD_h} \geq q_{O_kD_h}^{rail} \geq \sum_{S_j \in \mathbb{S}} \phi_{S_j}^{O_kD_h}, \forall k, h \in \mathscr{K}, h \neq k \tag{19}$$

$$\sum_{S_j \in \mathbb{S}} \phi_{S_j}^{O_kD_h} \leq \mathbf{1} \tag{20}$$

(5) Constraints of the decision variables' values

$$p_{ij,\min}^{rail} \leq p_{ij}^{rail} \leq p_{ij,\max}^{rail}, \ \forall i \in \mathbb{N}, j \in \mathscr{D} \tag{21}$$

$$q_{ij}^{rail} \geq 0, \ q_{ij}^{road} \geq 0, \ \forall i \in \mathbb{N}, j \in \mathscr{D} \tag{22}$$

$$\phi_{S_j}^{O_kD_h} \in \{0,1\}, \psi^k, m_k \in Z, \quad \forall S_j \in \mathbb{S}, k, h \in \mathscr{K}, h \neq k \tag{23}$$

3.4. Lower level: Network equilibrium model

The utility function of each transportation mode can be expressed as follows:

$$V_{ij}^m = -\alpha_1^m p_{ij}^m - \alpha_2^m t_{ij}^m - \alpha_3^m r_{ij}^m + \alpha_4^m v_{ij}^m, \forall m \in \mathscr{M} \tag{24}$$

where p_{ij}^m, t_{ij}^m, r_{ij}^m, and v_{ij}^m indicate freight rate, transportation time, carbon tax, and mode convenience, respectively. $\alpha_i^m (i = 1,2,3,4)$ is the weight of these factors.

Then the generalized cost of the mth $(m \in \mathscr{M})$ transportation mode is represented herein as a power function.

$$f_{ij}^m(x) = a\left(q_{ij}^m\right)^b - V_{ij}^m \tag{25}$$

where a and b are pending parameters.

Consigners choose the transportation mode with a minimal general cost. The freight demand for each mode will eventually reach a stable equilibrium. The market share under this equilibrium can be expressed by the following mathematical model (Jin, 2005):

$$\mathbf{min} \sum_{(i,j)\in\mathscr{L}}\sum_{m\in\mathscr{M}} \int_0^{q_{ij}^m} f_{ij}^m(x)\ d(x) \tag{26}$$

$$\mathbf{s.t.} \sum_{m\in\mathscr{M}} q_{ij}^m = B_{ij}, \ \forall i \in \mathbb{N}, j \in \mathscr{D} \tag{27}$$

$$q_{ij}^m \geq 0, \forall m \in \mathscr{M}, \ i \in \mathbb{N}, j \in \mathscr{D} \tag{28}$$

4. Algorithm design

A bi-level optimization model's optimal solution is computationally intractable due to the difficulties in evaluating the upper-level objective function and the interaction between the two levels. The proposed model is even more challenging since it

considers multi objectives at the upper-level in addition to the bi-level structure. In this paper, we design a hybrid solution procedure that combining NSGA-III and SAB to solve the bi-level multi-objective model. Genetic algorithm is one of the most commonly used heuristics. For example, Patidar et al. (2018) used a genetic algorithm to solve a sustainable vehicle routing problem in the design of an Indian agricultural food supply chain network. Pratap et al. (2016) combined a genetic algorithm, a greedy heuristic, and an advanced block-based evolutionary algorithm to solve an integrated problem of optimizing the operations at a commercial bulk material port terminal. NSGA-III is a variant of genetic algorithm in the field of multi-objective optimization.

1) Initialization

Set the initial freight rates for railway transportation and road transportation as $(p_{ij}^{rail(ite)}, p_{ij}^{road(ite)})$. Set $ite = 0$.

2) The lower-level model solving

Substitute the freight rates $(p_{ij}^{rail(ite)}, p_{ij}^{road(ite)})$ into the lower-level model. Then CVX toolbox of MATLAB is used to get the equilibrium freight volume $(q_{ij}^{rail(ite)}, q_{ij}^{road(ite)})$.

The key to solving a bi-level programming problem is to find the specific relationship called response function between the lower-level decision variables and the upper-level decision variables. When there are multiple modes in a transportation system, the sensitivity analysis can obtain the derivative relationship between the share of an individual transportation mode and its transportation price. Then the linear approximation of the reaction function can be made by using the Taylor expansion.

Assuming that $u_{ij}^{rail(0)}$ is the initial rail freight price from station $i \in \mathbb{N}$ to station $j \in \mathscr{D}$ and the road freight price remains unchanged. The equilibrium rail freight volume $q^{rail}(u_{ij}^{rail(0)})$ is obtained by solving the lower-level model. Then by using the proposed sensitivity analysis, the derivative relationship $\partial q^{rail}/\partial P_{ij}^{rail}$ between the freight volume $q^{rail}(P_{ij}^{rail})$ and the freight price P_{ij}^{rail} of rail transportation is achieved. The reaction function is approximated by Taylor expansion:

$$q^{rail}\left(P_{ij}^{rail}\right) = q^{rail}\left(u_{ij}^{rail(0)}\right) + \frac{\partial q^{rail}}{\partial P_{ij}^{rail}}\left(P_{ij}^{rail} - u_{ij}^{rail(0)}\right) \tag{29}$$

Rearrange the equation (29) as:

$$P_{ij}^{rail}\left(q^{rail}\right) = u_{ij}^{rail(0)} + \frac{\partial P_{ij}^{rail}}{\partial q^{rail}}\left(q^{rail} - q^{rail}\left(u_{ij}^{rail(0)}\right)\right) \tag{30}$$

3) The upper-level model solving

The linear approximation of the reaction function is substituted into the upper-level model. Then solve the upper-level model by NSGA-III (the details of NSGA-III is described as Fig. 3) to get $(q_{ij,k}^{rail(ite)}, \phi_{S_i}^{O_k D_h(ite)}, \psi^{k(ite)}, m^{k(ite)})$. The new rail freight price $p_{ij}^{rail(ite+1)}$ is obtained by substituting the railway freight volume $q_{ij,k}^{rail(ite)}$ into the equation (30).

a) *Objective function normalization*

The first step is to define the ideal points of the population S_k as the minimum value $z^{\min} = (z_1^{\min}, z_2^{\min}, \cdots, z_M^{\min})$ of $\cup_{t=0}^{k} S_k$, and then the objectives function can be converted to $f_j'(x) = f_j(x) - z_j^{\min}$. The second step is to find the extreme points corresponding to each coordinate axis, then use them to construct a hyperplane and find the intercept a_j between the hyperplane and the coordinate axis.

$$ASF(x, w) = max_{i=1}^{M} f_j'(x) \Big/ w_i,\ x \in S_k \tag{31}$$

$$z^{i,\max} = s : argmin_{s \in S_k} ASF\left(x, w^i\right), w^i$$

$$= \left(10^{-6}, 10^{-6}, \cdots, 10^{-6}\right), w_j^i = 1 \tag{32}$$

Finally, the objectives can be adaptively normalized as follows:

$$f_j^{norm}(x) = \frac{f_j(x) - z_j^{\min}}{a_j - z_j^{\min}}, j = 1, 2, \cdots, M \tag{33}$$

b) *Association*

By calculating perpendicular distance from members to reference lines (joining the ideal point with a reference point, the method of creating reference points is based on Das and Dennis's systematic approach (Das and Dennis, 1998)), each individual will be associated with a reference point having the minimum perpendicular distance.

c) *Niche-Preservation operation*

Select a reference point with the fewest associated individuals and denote the number of associated individuals as ρ_j. If $\rho_j = 0$ and there is an individual associated with the reference point in the current non-dominated layer, then pick one associated individual with the shortest perpendicular distance to join the next generation population. If no individual is selected, the reference point is removed from the current generation. If $\rho_j > 0$, a randomly chosen individual associated with the reference point in the current non-dominated layer will be added to the next generation population. Repeat Niche-Preservation operation until $|P_{k+1}| = N$.

4) Stopping criterion

Repeat 2) and 3) until $\left|p_{ij}^{rail(ite+1)} - p_{ij}^{rail(ite)}\right| \leq \omega$, where ω is the relative error.

5. Case study

5.1. The input data

In this section, a transportation network with nine nodes shown in Fig. 4 is designed to verify the proposed methodology. Three types of trains are operating in this network, and their routes are shown in Table 3.

The transportation mileage, transportation time, current freight price, and freight demand of the 18 OD pairs are shown in Table 4. These data are derived from China Railway Corporation (2019) and Alibaba Logistics (2019). The necessary parameter values required in the model are shown in Table 5.

5.2. Results and discussion

Four different scenarios are designed in this section. Table 6 shows the different objective functions and considerations of them.

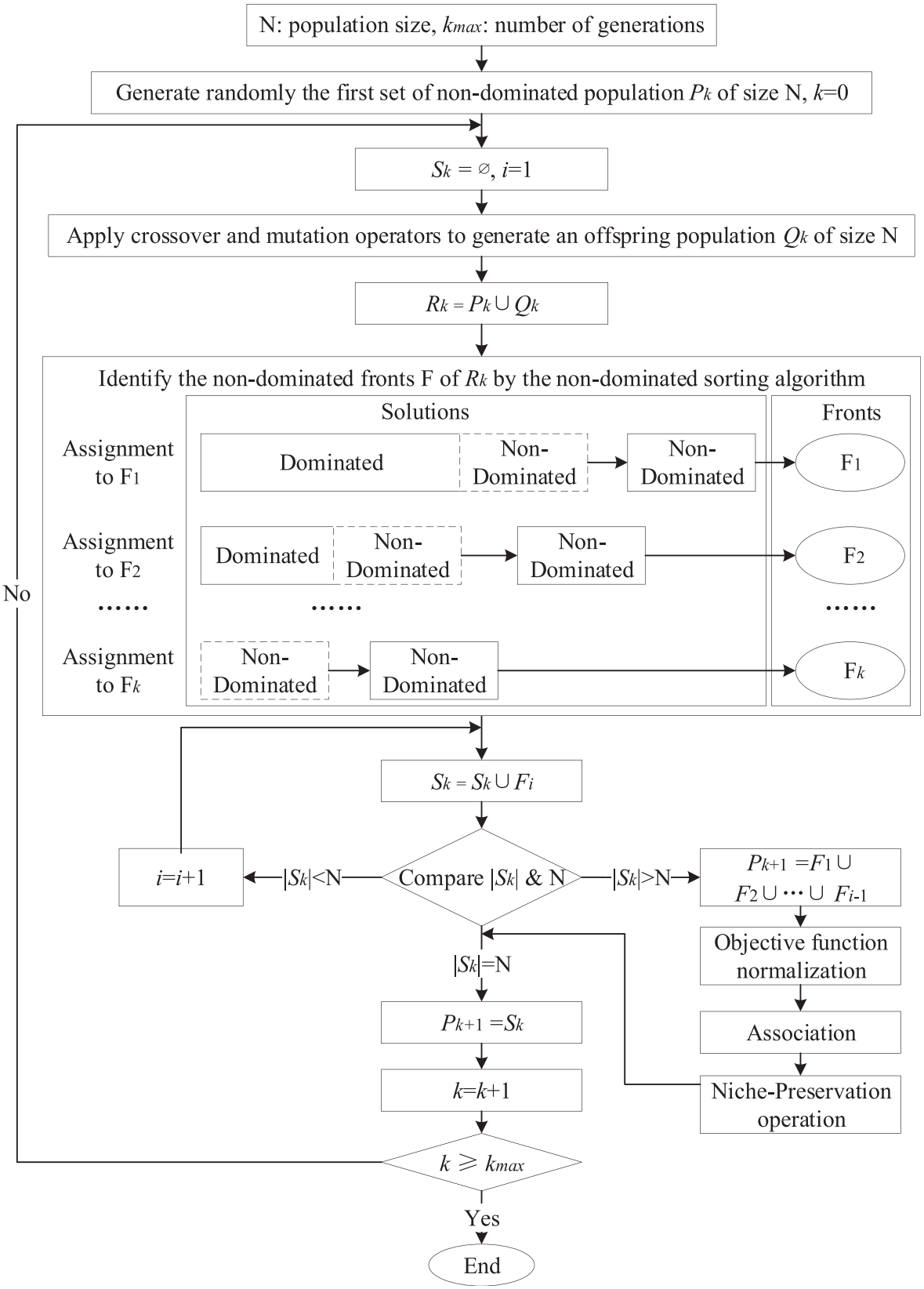


Fig. 3. Flowchart of NSGA-III.



Fig. 4. Rail transportation network with nine nodes in China. The map of China comes from the National Platform for Common Geospatial Information Services developed by the Ministry of Natural Resources of China (MNR, 2020).

5.2.1. Scenario #0: Reference

This paper does not consider other freight modes except road transportation and rail transportation. According to the freight volume of China in 2018 (NBS, 2019), it can be calculated that the market share of road freight is 90.76% and that of rail freight is 9.24%. This mode split is used as the reference scenario. In this scenario, 891 tons of cargos are transported by rail and 8754 tons of cargos are transported by road.

According to equations (1)–(3), we can calculate that 949.434 tons of carbon emissions will be generated if all the freight demands are met by road transportation. In this scenario, the carbon emissions generated are 882.910 tons. The carbon emission reduction rate is 7.01% compared to the case of road-only.

5.2.2. Scenario #1: Cost-only

Carbon emission reduction policies are ignored in this scenario. Rail freight operators make the operation planning and pricing decisions solely based on economic performance. The environmental cost is not taken into account when consigners choose the mode of transport, i.e., $\alpha_3 = 0$ is set.

Table 3
Train route.

Train	Route	Notation
K_1	Beijing-Zhengzhou-Wuhan-Changsha-Nanning	$O_1 - S_1 - S_2 - S_3 - D_1$
K_2	Jinan-Zhengzhou- Wuhan-Changsha-Kunming	$O_2 - S_1 - S_2 - S_3 - D_2$
K_3	Xi'an-Zhengzhou- Wuhan-Changsha-Guangzhou	$O_3 - S_1 - S_2 - S_3 - D_3$

Table 4
Mileage, freight time, current freight price and freight demand on various ODs.

OD pairs	Rail mileage (km)	Road mileage (km)	Time spent on rail transport (day)	Time spent on road transport (day)	Current rail freight rate (yuan/ton)	Current road freight price (yuan/ton)	Daily freight demand (ton)
$O_1 \rightarrow D_1$	2567	2657	7	4	544.8	615.1	720
$O_1 \rightarrow D_2$	3184	3228	8	4	558.2	747.3	900
$O_1 \rightarrow D_3$	2295	2478	6	3	467.4	573.6	360
$O_2 \rightarrow D_1$	2538	2297	7	4	487.8	531.8	540
$O_2 \rightarrow D_2$	3119	2907	8	4	569	672.9	755
$O_2 \rightarrow D_3$	2151	2027	6	3	410.5	469.3	360
$O_3 \rightarrow D_1$	2383	2073	6	5	388.1	479.8	414
$O_3 \rightarrow D_2$	1942	2004	5	5	370.3	463.9	450
$O_3 \rightarrow D_3$	2111	2033	6	5	409	470.6	990
$S_1 \rightarrow D_1$	1872	1935	5	4	372.3	447.9	395
$S_1 \rightarrow D_2$	2489	2546	6	4	431.8	589.3	558
$S_1 \rightarrow D_3$	1600	1756	5	3	323.7	406.5	540
$S_2 \rightarrow D_1$	1336	1404	4	3	290.2	325.0	270
$S_2 \rightarrow D_2$	1953	2014	5	3	381.4	466.2	450
$S_2 \rightarrow D_3$	1064	1225	4	2	236.3	283.6	414
$S_3 \rightarrow D_1$	978	1012	3	3	217.6	234.3	540
$S_3 \rightarrow D_2$	1595	1622	5	4	308.8	375.5	270
$S_3 \rightarrow D_3$	706	833	3	2	163.7	192.8	720

Table 5
Numerical parameter values employed in the empirical case study.

Parameters	Values	Data Source
$[\sigma^{rail}, \sigma^{road}]$ (ton/ton-km)	$[11.4 \times 10^{-6}, 47.1 \times 10^{-6}]$	Lin et al. (2017)
G_{teu} (ton/TEU)	21.6	China Railway Corporation (2019)
c_{fix}^{k} (yuan/train)	[77010,93570,63330]	
c_{var}^{k} (yuan/TEU)	[963,1170,792]	
$c_{transfer}$ (yuan/ton)	126	
$R_{k}^{\max}$ (car)	60	
$q_{i}^{\max}$ (ton)	7000	
ξ_{w} (train)	6	
$\rho_{\min}$	80%	
cv_{ij}^{m}	[0,25]	Zhang and Li (2019)
$[a, b]$	[3,0.3]	
$[\alpha_1, \alpha_2, \alpha_3, \alpha_4]$	[0.3, 0.4, 0.15, 0.15]	Calculated by using the analytic hierarchy process of the survey data

Table 6
The objective functions and considerations in **Scenario #1-3**.

Scenario	The objectives of the upper-level model	The composition of generalized transportation costs of consigners in the lower-level model
Scenario #1	$\max I(\boldsymbol{p}, \boldsymbol{q}) - C(\boldsymbol{q}, \psi, \boldsymbol{\tau}, \boldsymbol{\phi})$	Freight time, price, and convenience
Scenario #2	$\max I(\boldsymbol{p}, \boldsymbol{q}) - C(\boldsymbol{q}, \psi, \boldsymbol{\tau}, \boldsymbol{\phi})$ **max** $\mu(\boldsymbol{q})$	Freight time, price, convenience, and carbon tax
Scenario #3	$\max I(\boldsymbol{p}, \boldsymbol{q}) - C(\boldsymbol{q}, \psi, \boldsymbol{\tau}, \boldsymbol{\phi}) + U(\boldsymbol{q})$ **max** $\mu(\boldsymbol{q})$ min $U(\boldsymbol{q})$	Freight time, price, convenience, and carbon tax

5.2.3. Scenario #2: External costs added

Based on **Scenario #1**, a government-developed carbon tax policy acting on the demand-side (consignors) is implemented in this scenario (set $\eta = 600$). It is a way to internalize environmental costs into the transportation costs of the consigners.

5.2.4. Scenario #3: Subsidies added

Based on **Scenario #2**, in this scenario, not only the carbon tax policy is implemented but also a low-carbon subsidy policy that acts on the supply-side (rail freight operators) is performed (set $\eta = 600$, $\lambda = 1000$).

The upper-level models in **Scenario #2-3** are multi-objective. In **Scenario #3**, a Pareto frontier with 80 solutions shown in Fig. 5 can be obtained according to the methodology proposed above. The algorithm is conducted with MATLAB R2018a on a 1.80 GHz Intel Core laptop, with a total memory of 8 GB.

For expositional reasons, and also to understand the solutions contained within each point depicted in Fig. 5, we focus on the maximum profit solution. Tables 7–12 provides the relevant details on the rail freight train services of **Scenario #1-3**.

Tables 7 and 8 respectively show the optimal rail freight price and freight volume in each OD for **Scenario #1-3**. Note that not all cargos can be transported to the destination directly by one train. For example, train K_1 with the terminal D_1 is also loaded with cargos that need to be transported to D_2 and D_3 at the starting station O_1. These two parts of cargos must be transshipped at one of the three intermediate stations (S_1, S_2, S_3).

Cargos that cannot be directly transported to the destination can be transshipped according to the schemes in Tables 9 and 10. For example, in **Scenario #1**, the cargos that need to be transported to D_3 on train K_1 need to be unloaded at S_3, then taken by train K_3 at S_3 and eventually transported to D_3.

The optimal rail freight service frequency and number of cars in each train are shown in Table 11. It satisfies the constraint that the

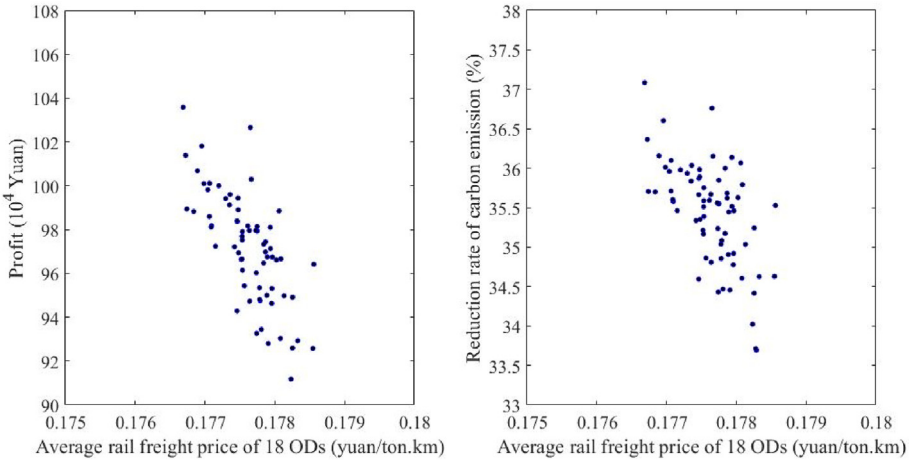


Fig. 5. The Pareto frontier of **Scenario #3**.

Table 7
Optimal rail freight price (yuan/ton) for **Scenario #1-3**.

OD pairs	Scenario #1	Scenario #2	Scenario #3
$O_1 \rightarrow D_1$	486.1	456.2	451.6
$O_1 \rightarrow D_2$	527.4	638.3	616.6
$O_1 \rightarrow D_3$	391.9	435.0	472.1
$O_2 \rightarrow D_1$	411.7	369.0	363.6
$O_2 \rightarrow D_2$	433.6	447.4	511.8
$O_2 \rightarrow D_3$	302.8	335.4	367.6
$O_3 \rightarrow D_1$	394.5	419.9	430.3
$O_3 \rightarrow D_2$	407.5	429.5	441.7
$O_3 \rightarrow D_3$	418.0	402.4	497.5
$S_1 \rightarrow D_1$	314.8	304.4	332.9
$S_1 \rightarrow D_2$	454.2	484.5	439.6
$S_1 \rightarrow D_3$	264.0	230.8	264.5
$S_2 \rightarrow D_1$	218.7	218.3	209.5
$S_2 \rightarrow D_2$	287.9	330.1	290.8
$S_2 \rightarrow D_3$	191.2	184.1	176.5
$S_3 \rightarrow D_1$	214.6	181.7	177.2
$S_3 \rightarrow D_2$	220.7	267.8	268.2
$S_3 \rightarrow D_3$	156.4	143.1	124.0

Table 8
Optimal rail freight volume (ton) for **Scenario #1-3**.

OD pairs	Scenario #1			Scenario #2			Scenario #3		
	K_1	K_2	K_3	K_1	K_2	K_3	K_1	K_2	K_3
$O_1 \rightarrow D_1$	215.9	0	0	394.4	0	0	450.2	0	0
$O_1 \rightarrow D_2$	360.6	0	0	192.4	0	0	490.0	0	0
$O_1 \rightarrow D_3$	121.0	0	0	145.3	0	0	168.5	0	0
$O_2 \rightarrow D_1$	0	93.8	0	0	179.3	0	0	212.0	0
$O_2 \rightarrow D_2$	0	328.5	0	0	388.6	0	0	488.6	0
$O_2 \rightarrow D_3$	0	157.1	0	0	103.1	0	0	130.1	0
$O_3 \rightarrow D_1$	0	0	209.2	0	0	121.7	0	0	175.2
$O_3 \rightarrow D_2$	0	0	170.5	0	0	295.9	0	0	208.5
$O_3 \rightarrow D_3$	0	0	261.5	0	0	388.6	0	0	178.8
$S_1 \rightarrow D_1$	235.5	0	0	359.5	0	0	212.1	0	0
$S_1 \rightarrow D_2$	0	227.5	0	0	308.8	0	0	224.0	0
$S_1 \rightarrow D_3$	0	0	231.3	0	0	237.0	0	0	381.4
$S_2 \rightarrow D_1$	72.4	0	0	146.2	0	0.0	161.7	0	0
$S_2 \rightarrow D_2$	0	153.0	0	0	167.3	0.0	0	222.6	0
$S_2 \rightarrow D_3$	0	0	71.7	0	0	152.6	0	0	120.2
$S_3 \rightarrow D_1$	249.5	0	0	231.7	0	0	323.3	0	0
$S_3 \rightarrow D_2$	0	140.9	0	0	96.2	0	0	112.8	0
$S_3 \rightarrow D_3$	0	0	344.9	0	0	219.3	0	0	418.5
Sum	3645			4128			4679		

load factor is not less than 80%, which is beneficial to the control of transportation costs. It can be found from Table 11 that the number of cars in **Scenario #3** is higher than that in **Scenario #1** and **2**, which is consistent with the result shown in Table 8. According to Table 8, the integrated optimization of operation planning and pricing strategy in **Scenario #1** can increase the rail freight market share by 28.55% (compared with **Scenario #0**). The carbon tax policy in **Scenario #2** can increase the rail freight market share by another 5.01% (compared with **Scenario #1**). The low-carbon subsidy policy in **Scenario #3** can increase the rail freight market share by another 5.71% (compared with **Scenario #2**).

MS potential from road to rail, carbon emission reduction rate, and benefits/costs of the three stakeholders (railway freight operators, the government, and consigners) are calculated. The results are shown in Table 12, which indicate that the approach proposed in this paper that combining the improvement of railway freight service level and the implementation of carbon emission reduction policies can achieve a 39.27% MS from road to rail, as well as a 37.1% carbon emission reduction in the service chain.

Compared with **Scenario #1**, the implementation of the carbon tax policy in **Scenario #2** can increase the carbon emission

Table 9
Optimal unloading operation plan in intermediate stations for **Scenario #1-3**.

Scenario	Train	Unloading operation		
		S_1	S_2	S_3
Scenario #1	K_1	–	–	O_1D_2, O_1D_3
	K_2	–	O_2D_1	O_2D_3
	K_3	O_3D_1	–	O_3D_2
Scenario #2	K_1	O_1D_3	–	O_1D_2
	K_2	–	O_2D_1, O_2D_3	–
	K_3	–	O_3D_1, O_3D_2	–
Scenario #3	K_1	–	–	O_1D_2, O_1D_3
	K_2	–	O_2D_1	O_2D_3
	K_3	O_3D_1	–	O_3D_2

Table 10
Optimal loading operation plan in intermediate stations for **Scenario #1-3**.

Scenario	Train	Loading operation		
		S_1	S_2	S_3
Scenario #1	K_1	O_3D_1	O_2D_1	–
	K_2	–	–	O_1D_2, O_3D_2
	K_3	–	–	O_1D_3, O_2D_3
Scenario #2	K_1	–	O_2D_1, O_3D_1	–
	K_2	–	O_3D_2	O_1D_2
	K_3	O_1D_3	O_2D_3	–
Scenario #3	K_1	O_3D_1	O_2D_1	–
	K_2	–	–	O_1D_2, O_3D_2
	K_3	–	–	O_1D_3, O_2D_3

Table 11
Optimal rail train operating plan for **Scenario #1-3**.

Operating plan	Scenario #1			Scenario #2			Scenario #3		
	K_1	K_2	K_3	K_1	K_2	K_3	K_1	K_2	K_3
Service frequency	1	1	1	1	1	1	1	1	1
Number of cars	25	32	28	34	38	32	44	45	33

reduction rate by 4.07% and increase the railway operators' profits by 44.18%. Compared with **Scenario #2**, the implementation of the carbon emission subsidy policy in **Scenario #3** can increase the carbon emission reduction rate by 4.65%. However, the railway operators' profits (excluding the subsidies received) do not improve significantly compared to **Scenario #2**, which may be due to lower freight rates in order to attract more freight demand. But subsidies paid by the government can fill this gap, which shows that the carbon subsidy policy is helpful to improve the enthusiasm of railway transportation.

Pricing plays a crucial role in the restrictive relationship among the three stakeholders. Through subsidies, rail operators can provide more competitive rate than roads. The freight rate provided directly affects consigners' mode choice, and the result determines whether the government's carbon emission reduction target can be achieved. For railway operators, the joint optimization of operation management and revenue management is more conducive to balancing the relationship between transport supply and demand than the separate optimization of operation planning or pricing, which can improve the utilization of railway capacity. The significance of the proposed methodology for MS from road to rail is even better than that of external policies. For the government, the policy package has a better carbon emission reduction effect than a single policy and will become the development trend of future carbon emission reduction practices. Besides, policymakers should not formulate carbon policies based solely on the mitigation effects of individual OD pairs. Instead, Policymakers should assess overall

Table 12
Summary of simulation outputs.

Scenario	MS from road to rail (ton, compared with Scenario #0)	Carbon emissions (ton)	Reduction rate of carbon emissions	Profits of rail freight operators (yuan)	The environmental costs paid by consigners to rail and road respectively (yuan)	Subsidies provided by the government to rail (yuan)
Scenario #1	2754 (28.55%)	680.050	28.37%	648959.9	–	–
Scenario #2	3237 (33.56%)	641.393	32.44%	935682.0	[58634.2, 326201.2]	–
Scenario #3	3788 (39.27%)	597.321	37.09%	1035875.9	[66890.7, 291501.9]	111484.4

emissions reductions in the transportation network.

5.3. Variation in carbon tax and subsidy

In order to study the impact of the carbon tax rate that is set at 600 yuan per ton carbon emissions in **Scenario #3**, we considered five additional settings: $\eta = 0$, $\eta = 300$, $\eta = 900$, $\eta = 1200$, and $\eta = 1500$, while keeping the carbon subsidy parameter unchanged. Fig. 6 provides a comparative snapshot of the different settings. It is found that when the carbon tax policy is not implemented, the rail operators do not occupy a high market share even if it sets a low freight rate, and its profit is still low. This is because the railway advantage of low-carbon environmental protection has not played a role in this case. After internalizing the environmental costs into the transportation costs of the consigners, that is, setting an appropriate carbon tax, the rail operators will obtain higher profits by attracting more freight demand. However, excessive carbon tax will have an inhibiting effect on consigners' willingness to transport. In Fig. 6, the profits of railway operators decrease when the carbon tax rate exceeds 900, which is not conducive to carbon emission reduction. In this case, railway operators can lower the freight price appropriately to attract more demand.

While maintaining the carbon tax rate, the effect of the carbon emission subsidy coefficient on the solution is studied by setting $\lambda = [0, 500, 1000, 1500, 2000, 2500]$ yuan per ton carbon emissions. The relevant solution highlights are depicted in Fig. 7, indicating that higher carbon subsidy has made the railway more profitable. However, when the carbon emission subsidy factor is too large, the marginal cost of the government is more considerable. That is, the government pays a more substantial subsidy for each additional unit of railway traffic. Based on the government's expectation of the least subsidy, a large carbon subsidy factor may lead the government to choose a scheme with a relatively small market share in railway transportation where feasible. At the same time, the railway sector receives more subsidies per unit turnover, which may eliminate the need for higher volumes for more profits. As shown in Fig. 7, the total profits obtained by the railway at the subsidy coefficient of 2500 is relatively equal to that at the subsidy coefficient of 2000, indicating that the transportation revenue of the railway decreases, but the high subsidy fills the gap. Therefore, the carbon reduction rate is determined by the balance of interests between the government and the railway sector, but it is not more favorable to reduce carbon emissions through greater subsidies.

Through the above analysis, we can get the following views. Policymakers should design their carbon policies carefully to achieve the balance between benefiting the environment and satisfying customers. A carbon tax in the middle range might be preferable. However, a high carbon tax policy will lead to excessively high transportation costs paid by customers, which in turn will reduce customers' sensitivity to carbon emissions penalties and is not good for increasing demand for low-carbon products. The advantage of subsidy policy tools is that, unlike tax policy tools, they can increase the market share of rail transport without affecting transport costs. But they force governments to increase fiscal spending. The government must fully consider the interests of transportation companies to ensure the effective implementation of carbon reduction policies. However, excessive subsidies to the rail freight system must be minimized to prevent carrier inertia. Determining how to comprehensively consider social costs to achieve emission reduction goals with minimal government investment is a key issue that governments must consider.

6. Conclusion

This research proposes a decarbonizing methodology for an intraregional freight system, which focuses on promoting MS from

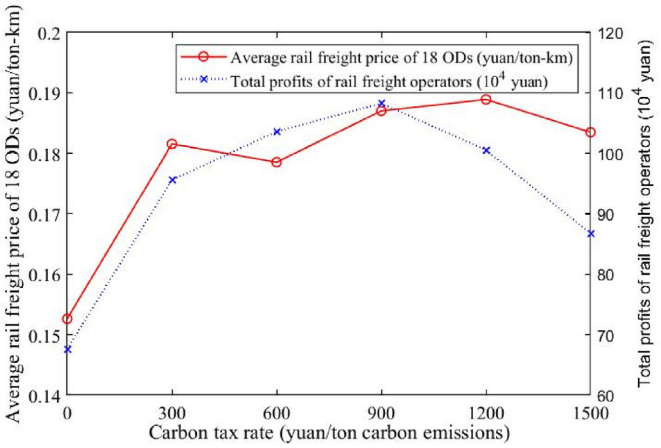


Fig. 6. Impact of the carbon tax policy.

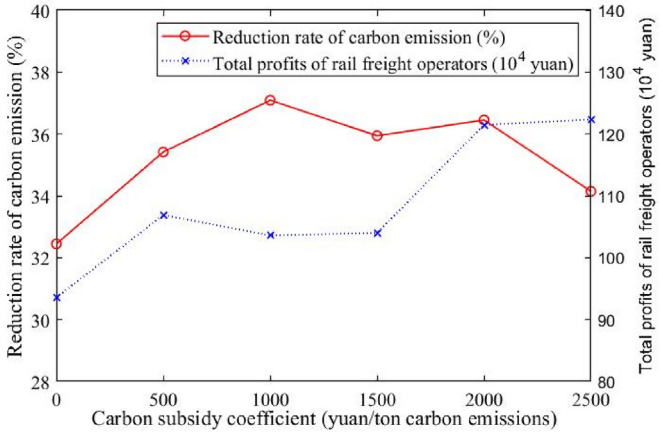


Fig. 7. Impact of subsidy policy.

road to rail. The trade-off among three parties (rail operators, consigners, and the government), improvement of rail freight service quality, mode choice behavior, and effects of tax and subsidy policy tools on carbon emission reduction, are comprehensively considered to establish a programming model based on the bi-level multi-objective optimization. It is extremely tough to get the optimal solution for this problem. However, a hybrid algorithm combining NSGA-III and SAB has the potential to find near-optimal railway operating plans and prices.

The numerical case analysis based on a China freight network suggests that the joint optimization of operation plan and freight rate in a free market competition environment is of great significance for promoting MS. Pricing plays a bridge role in the balance among the three stakeholders, and jointly optimizing it with train operation decisions is conducive to achieving the optimal capacity allocation for railway transportation. The improvement of the railway's service quality is the most effective and feasible way to achieve its high market share. Additional carbon tax policy and low-carbon subsidy policy can further promote MS, but the government must formulate appropriate standards. Compared with **Scenario #1** that without government intervention, the implementation of the carbon tax policy in **Scenario #2** can reduce the carbon emissions of the transportation system by another 4.0%, and the joint implementation of the carbon tax and carbon subsidy policies in **Scenario #3** can reduce the carbon emissions by another 8.7%. It can be seen that the application of multiple policies is more useful than a single policy. Higher carbon taxes and subsidies are not necessarily better for reducing carbon emissions. In the case study of this paper, the carbon emission reduction rate is the maximum when $(\eta, \lambda) = (600, 1000)$ yuan per ton of carbon emissions is set. Under this policy combination, the rail freight will achieve a high market share, consigners will not avoid freight due to excessive carbon tax, and the government will not be subject to excessive economic pressure.

Future researches can be focused on the following issues. First, an extended model of the present work can be developed by integrating the rail and road modes into an intermodal network. Second, a non-linear demand function can be considered for model development due to the demand elasticity resulting out of shippers' mode choice behavior. Third, how the government formulates the optimal carbon tax rate and carbon subsidy intensity based on a preset emission reduction target deserves further study.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Lin Li: Conceptualization, Methodology, Software, Formal analysis, Writing - original draft, Visualization, Data curation, Funding acquisition. **Xiaoqiang Zhang:** Validation, Investigation, Resources, Writing - review & editing, Supervision, Project administration, Funding acquisition.

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